

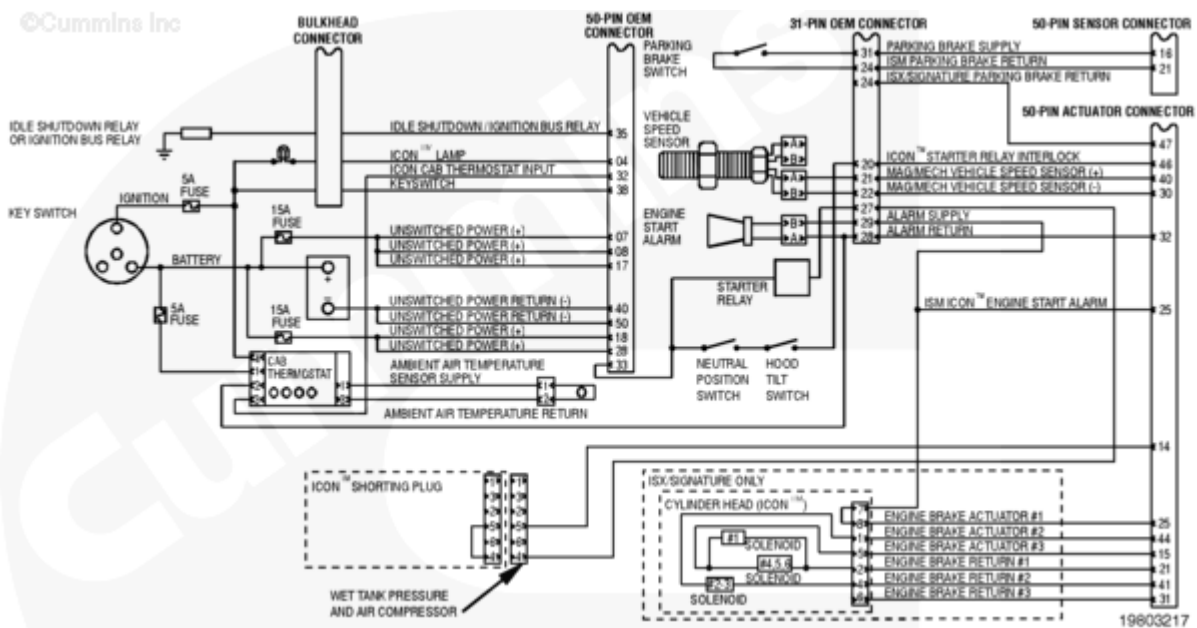
Fault Code: 359 (Integrated)

Engine Failed to Start (Automatic Start) - Condition Exists

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Overview

Codes	Reason	Effect
Fault Code: 359 PID(P): SPN: FMI: 11 Lamp: Yellow SRT:	Engine Failed to Start (Automatic Start) - Condition Exists. The ICON™ system has failed to start the engine automatically.	The ICON™ system will be disabled. Only mandatory shutdown will be enabled. Can possibly start engine normally.



Circuit Description

The starter relay circuit controls and monitors both the starter relay coil and return signal. The starter relay is used by the ICON™ feature to perform automatic starts of the engine.

Component Location

The starter relay is mounted on the fire wall of the vehicle on the intake side of the engine.

Shoptalk

This fault code is set if two consecutive automatic starts fail. If a start is commanded by the engine electronic control module (ECM) and 200 rpm is **not** reached within 2 seconds nor 450 rpm within 14 seconds, then the start failed. After the first failure, the ICON™ system waits 1 minute and tries again. If the second start fails, the fault is set. It is cleared as soon as a manual start is successful.

Warnings and Cautions

CAUTION

To reduce the possibility of damaging a new engine ECM, all other active fault codes must be investigated prior to replacing the engine ECM.

To reduce the possibility of pin and harness damage, use the following test leads when taking a measurement:

Part Number 3822917 - female Deutsch/AMP/Metri-Pack test lead

Part Number 3822758 - male Deutsch/AMP/Metri-Pack test lead.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Interview the driver to determine when the failure occurred, what the general setup information is, and if any hard start problems occur during manual starts.	
	STEP 1A. Interview the driver to determine when the failure occurred, what the general setup information is, and if any hard start problems occur during manual starts.	Engine problem
STEP 2.	Read all fault codes.	
	STEP 2A. Read fault codes with INSITE™ electronic service tool or flash out with ICON™ lamp.	Fault Code 359 inactive

STEPS		SPECIFICATIONS
	STEP 2B. Manually start the engine.	Engine starts
STEP 3.	Check the starter relay contacts.	
	STEP 3A. Perform the starter relay and interlocks test.	Starter engages
	STEP 3B. Check for an open circuit in the ICON™ starter relay coil.	Less than 10 ohms
STEP 4.	Activate the ICON™ system; check the engine ECM for active fault codes.	
	STEP 4A. Activate the ICON™ system; check the engine ECM for active fault codes.	No active fault codes; Engine performs autostart
STEP 5.	Clear the fault codes.	
	STEP 5A. Disable the fault code.	Fault Code 359 inactive

STEP 1. Interview the driver to determine when the failure occurred, what the general setup information is, and if any hard start problems occur during manual starts.

STEP 1A. Interview the driver to determine when the failure occurred, what the general setup information is, and if any hard start problems occur during manual starts.

Conditions :

- Check the settings of the cab-mounted thermostat for ICON™ system.
- Make certain that the idle shutdown timer and cab thermostat settings are correct in INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
	Engine problem	Appropriate troubleshooting and repair manual
	Fault Code 359, ICON™ system problem	2A

STEP 2. Read all fault codes.

STEP 2A. Read fault codes with INSITE™ electronic service tool or flash out with ICON™ lamp.

Conditions :

- Connect all components.
- Turn keyswitch ON.

Action	Specification/Repair	Next Step
• Read the fault codes with INSITE™ electronic service tool or flash out with ICON™ lamp.	Fault Code 359 inactive	5A
	Fault Code 359 active or Fault Code 2291 inactive	2B

STEP 2B. Manually start the engine.

Conditions :

- Connect all components.
- Turn keyswitch ON.

Action	Specification/Repair	Next Step
• Start the engine using the keyswitch.	Engine starts	3A
	Refer to the appropriate troubleshooting and repair manual for hard start symptoms.	Appropriate troubleshooting and repair manual

STEP 3. Check the starter relay contacts.

STEP 3A. Perform the starter relay and interlocks test.

Conditions :

- Connect the INSITE™ electronic service tool.
- Turn keyswitch ON.

Action	Specification/Repair	Next Step
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Perform the starter relay and interlocks test using INSITE™ electronic service tool. Refer to the appropriate electronic service tool manual.	Starter engages	4A
	Starter does not engage	3B

STEP 3B. Check for an open circuit in the ICON™ starter relay circuit.

Conditions :

- Turn keyswitch OFF.
- Disconnect the ICON™ starter relay.

Action	Specification/Repair	Next Step
Measure the resistance between pin 14 of the ECM actuator connector to pin 33 of the ECM OEM connector. Refer to the wiring diagram or the circuit description at the beginning of this fault code for connector pin identification. For general resistance measurement techniques, refer to the Resistance Measurements Using a Multimeter and Wiring Diagram, Procedure 019-360 (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html).	Less than 10 ohms	4A
	Does not meet specifications. Repair or replace the OEM harness Refer to the OEM service manual.	4A

STEP 4. Activate the ICON™ system; check the engine ECM for active fault codes.

STEP 4A. Activate the ICON™ system; check the engine ECM for active fault codes.

Conditions :

- Connect all components.
- Start and run the engine in ICON™ mode.
- Perform autostart with the cab thermostat or battery voltage request.

Action	Specification/Repair	Next Step
Perform an autostart with the ICON™ system. After initial activation of the ICON™ system, engine will shut down in 1 minute.	No active fault codes. Engine performs autostart	5A
Request autostart with the cab thermostat. If the cab thermostat is not available, load the batteries to less than 12.2 VDC.	Active fault codes present. Engine does not autostart	Troubleshoot active fault codes 5A

STEP 5. Clear the fault codes.

STEP 5A. Disable the fault code.

Conditions :

- Connect all components.
- Turn keyswitch ON.

Action	Specification/Repair	Next Step
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• Verify Fault Code 359 is inactive using INSITE™ electronic service tool. • Erase the inactive fault codes using INSITE™ electronic service tool.	Fault Code 359 inactive. All fault codes cleared.	Repair complete
	Return to the troubleshooting steps, or contact the nearest Cummins Authorized Repair Location if all the steps have been completed and rechecked. Troubleshooting any remaining active fault codes.	Appropriate troubleshooting charts

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